

Raport Ewaluacji Modeli Multi-Task Learning (MTL) dla algorytmu Random Forest

Zestaw Tematyczny: PROCESY ABSORPCJI (Caco2, Lipophilicity, Solubility, HIA, AMES)

Zadania MTL	Descriptors	Fingerprints	Embeddings
Caco2 + Lipophilicity	Caco2 RMSE: 0.4540 MAE: 0.3426 R²: 0.6755	Caco2 RMSE: 0.5702 MAE: 0.4430 R²: 0.4882	Caco2 RMSE: 0.6313 MAE: 0.4957 R²: 0.3714
	Lipophilicity RMSE: 0.8553 MAE: 0.6385 R²: 0.5049	Lipophilicity RMSE: 0.9004 MAE: 0.6912 R²: 0.4513	Lipophilicity RMSE: 0.9724 MAE: 0.7717 R²: 0.3600
Caco2 + Lipophilicity + Solubility	Caco2 RMSE: 0.4560 MAE: 0.3453 R²: 0.6726	Caco2 RMSE: 0.6002 MAE: 0.4592 R²: 0.4330	Caco2 RMSE: 0.6546 MAE: 0.4996 R²: 0.3240
	Lipophilicity RMSE: 0.8642 MAE: 0.6502 R²: 0.4945	Lipophilicity RMSE: 0.9364 MAE: 0.7265 R²: 0.4065	Lipophilicity RMSE: 1.0260 MAE: 0.8238 R²: 0.2876
	Solubility RMSE: 1.1085 MAE: 0.7583 R²: 0.7736	Solubility RMSE: 1.4835 MAE: 1.1358 R²: 0.5944	Solubility RMSE: 1.4324 MAE: 1.1104 R²: 0.6219
Caco2 + Lipophilicity + Solubility + HIA	Caco2 RMSE: 0.4539 MAE: 0.3443 R²: 0.6757	Caco2 RMSE: 0.5988 MAE: 0.4579 R²: 0.4356	Caco2 RMSE: 0.6540 MAE: 0.5003 R²: 0.3253
	Lipophilicity RMSE: 0.8607 MAE: 0.6479 R²: 0.4985	Lipophilicity RMSE: 0.9361 MAE: 0.7278 R²: 0.4069	Lipophilicity RMSE: 1.0258 MAE: 0.8241 R²: 0.2878
	Solubility RMSE: 1.1102 MAE: 0.7590 R²: 0.7729	Solubility RMSE: 1.4865 MAE: 1.1385 R²: 0.5928	Solubility RMSE: 1.4349 MAE: 1.1137 R²: 0.6206
	HIA Accuracy: 0.9138 F1: 0.9510 AUROC: 0.9577	HIA Accuracy: 0.8707 F1: 0.9282 AUROC: 0.8812	HIA Accuracy: 0.8448 F1: 0.9151 AUROC: 0.9550

Caco2 + HIA	Caco2 RMSE: 0.4432 MAE: 0.3310 R ² : 0.6907	Caco2 RMSE: 0.5159 MAE: 0.3989 R ² : 0.5811	Caco2 RMSE: 0.5537 MAE: 0.4339 R ² : 0.5164
	HIA Accuracy: 0.9138 F1: 0.9510 AUROC: 0.9718	HIA Accuracy: 0.8793 F1: 0.9327 AUROC: 0.9164	HIA Accuracy: 0.8621 F1: 0.9238 AUROC: 0.9848
Lipophilicity + Solubility	Lipophilicity RMSE: 0.8671 MAE: 0.6518 R ² : 0.4911	Lipophilicity RMSE: 0.9388 MAE: 0.7298 R ² : 0.4034	Lipophilicity RMSE: 1.0246 MAE: 0.8212 R ² : 0.2895
	Solubility RMSE: 1.1102 MAE: 0.7577 R ² : 0.7729	Solubility RMSE: 1.4842 MAE: 1.1379 R ² : 0.5941	Solubility RMSE: 1.4272 MAE: 1.1070 R ² : 0.6246
Caco2 + AMES	Caco2 RMSE: 0.4559 MAE: 0.3383 R ² : 0.6728	Caco2 RMSE: 0.5339 MAE: 0.4056 R ² : 0.5512	Caco2 RMSE: 0.5772 MAE: 0.4506 R ² : 0.4744
	AMES Accuracy: 0.7978 F1: 0.8160 AUROC: 0.8716	AMES Accuracy: 0.8198 F1: 0.8307 AUROC: 0.9024	AMES Accuracy: 0.8095 F1: 0.8285 AUROC: 0.8742
Caco2 + Lipophilicity + AMES	Caco2 RMSE: 0.4591 MAE: 0.3464 R ² : 0.6682	Caco2 RMSE: 0.5566 MAE: 0.4237 R ² : 0.5124	Caco2 RMSE: 0.6252 MAE: 0.4843 R ² : 0.3834
	Lipophilicity RMSE: 0.8591 MAE: 0.6433 R ² : 0.5005	Lipophilicity RMSE: 0.8990 MAE: 0.6890 R ² : 0.4529	Lipophilicity RMSE: 0.9970 MAE: 0.7940 R ² : 0.3272
	AMES Accuracy: 0.7916 F1: 0.8107 AUROC: 0.8694	AMES Accuracy: 0.8102 F1: 0.8212 AUROC: 0.8888	AMES Accuracy: 0.7999 F1: 0.8159 AUROC: 0.8707

Caco2 + Lipophilicity + Solubility + HIA + AMES	Caco2 RMSE: 0.4563 MAE: 0.3457 R²: 0.6723	Caco2 RMSE: 0.5949 MAE: 0.4538 R²: 0.4428	Caco2 RMSE: 0.6512 MAE: 0.4993 R²: 0.3311
	Lipophilicity RMSE: 0.8637 MAE: 0.6496 R²: 0.4951	Lipophilicity RMSE: 0.9411 MAE: 0.7337 R²: 0.4006	Lipophilicity RMSE: 1.0342 MAE: 0.8316 R²: 0.2761
	Solubility RMSE: 1.1101 MAE: 0.7578 R²: 0.7729	Solubility RMSE: 1.5005 MAE: 1.1482 R²: 0.5851	Solubility RMSE: 1.4523 MAE: 1.1273 R²: 0.6113
	HIA Accuracy: 0.9052 F1: 0.9463 AUROC: 0.9577	HIA Accuracy: 0.8707 F1: 0.9282 AUROC: 0.8855	HIA Accuracy: 0.8448 F1: 0.9151 AUROC: 0.9615
	AMES Accuracy: 0.7758 F1: 0.7952 AUROC: 0.8631	AMES Accuracy: 0.7641 F1: 0.7603 AUROC: 0.8682	AMES Accuracy: 0.7785 F1: 0.7821 AUROC: 0.8633

Raport Ewaluacji Modeli Multi-Task Learning (MTL) dla algorytmu Random Forest

Zestaw Tematyczny: PROCESY ELIMINACJI I METABOLIZMU (Half Life, Clearance, CYP3A4, VDss, AMES)

Zadania MTL	Descriptors	Fingerprints	Embeddings
Half Life + Clearance Hepatocyte	Half_Life RMSE: 111.6777 MAE: 27.8583 R²: 0.1305	Half_Life RMSE: 101.1292 MAE: 26.5738 R²: 0.2870	Half_Life RMSE: 113.3546 MAE: 32.5226 R²: 0.1042
	Clearance_Hepatocyte RMSE: 49.1557 MAE: 35.8531 R²: 0.0499	Clearance_Hepatocyte RMSE: 46.7611 MAE: 34.5948 R²: 0.1402	Clearance_Hepatocyte RMSE: 40.5360 MAE: 30.7331 R²: 0.3386
Half life + Clearance Hepatocyte + CYP3A4 Inhibition	Half_Life RMSE: 106.8522 MAE: 26.4415 R²: 0.2040	Half_Life RMSE: 103.1371 MAE: 27.0245 R²: 0.2584	Half_Life RMSE: 114.1644 MAE: 33.2589 R²: 0.0914
	Clearance_Hepatocyte RMSE: 49.8097 MAE: 36.6418 R²: 0.0244	Clearance_Hepatocyte RMSE: 47.1202 MAE: 36.1743 R²: 0.1269	Clearance_Hepatocyte RMSE: 42.0756 MAE: 33.8002 R²: 0.2874
	CYP3A4 Accuracy: 0.7510 F1: 0.6981 AUROC: 0.8273	CYP3A4 Accuracy: 0.7218 F1: 0.5487 AUROC: 0.8296	CYP3A4 Accuracy: 0.7275 F1: 0.5976 AUROC: 0.8117
Half life + Clearance Hepatocyte + CYP3A4 Inhibition + Vdss Lombardo	Half_Life RMSE: 104.9254 MAE: 25.6548 R²: 0.2325	Half_Life RMSE: 103.2801 MAE: 27.4125 R²: 0.2564	Half_Life RMSE: 114.3949 MAE: 32.4437 R²: 0.0877
	Clearance_Hepatocyte RMSE: 49.8197 MAE: 36.5718 R²: 0.0240	Clearance_Hepatocyte RMSE: 47.0796 MAE: 36.4629 R²: 0.1284	Clearance_Hepatocyte RMSE: 42.2479 MAE: 34.1191 R²: 0.2816
	VDss RMSE: 6.5870 MAE: 3.8174 R²: 0.0738	VDss RMSE: 9.3292 MAE: 2.9914 R²: 0.8595	VDss RMSE: 6.9946 MAE: 5.2797 R²: -0.0445
	CYP3A4 Accuracy: 0.7506 F1: 0.6981 AUROC: 0.8277	CYP3A4 Accuracy: 0.7174 F1: 0.5325 AUROC: 0.8243	CYP3A4 Accuracy: 0.7133 F1: 0.5600 AUROC: 0.8041

Clearance Hepatocyte + CYP3A4 Inhibition	Clearance_Hepatocyte RMSE: 49.7035 MAE: 36.3343 R ² : 0.0286 CYP3A4 Accuracy: 0.7490 F1: 0.6949 AUROC: 0.8304	Clearance_Hepatocyte RMSE: 47.3257 MAE: 36.0080 R ² : 0.1193 CYP3A4 Accuracy: 0.7324 F1: 0.5807 AUROC: 0.8358	Clearance_Hepatocyte RMSE: 41.7014 MAE: 32.9187 R ² : 0.3001 CYP3A4 Accuracy: 0.7441 F1: 0.6367 AUROC: 0.8260
Half Life + CYP3A4 Inhibition	Half_life RMSE: 105.0412 MAE: 25.9899 R ² : 0.2308 CYP3A4 Accuracy: 0.7376 F1: 0.6845 AUROC: 0.8189	Half_Life RMSE: 102.9759 MAE: 27.6032 R ² : 0.2608 CYP3A4 Accuracy: 0.6959 F1: 0.4444 AUROC: 0.8328	Half_Life RMSE: 115.3485 MAE: 35.0068 R ² : 0.0724 CYP3A4 Accuracy: 0.6983 F1: 0.5356 AUROC: 0.7862
Half life + AMES	Half_life RMSE: 109.0799 MAE: 27.1180 R ² : 0.1705 AMES Accuracy: 0.7854 F1: 0.8072 AUROC: 0.8668	N/A	Half_Life RMSE: 114.3801 MAE: 34.7669 R ² : 0.0879 AMES Accuracy: 0.7331 F1: 0.7747 AUROC: 0.8175
Half Life + Clearance Hepatocyte + CYP3A4 Inhibition + VDss + AMES	Half_Life RMSE: 107.1315 MAE: 26.4194 R ² : 0.1999 Clearance_Hepatocyte RMSE: 49.9772 MAE: 36.6946 R ² : 0.0179 VDss RMSE: 7.8516 MAE: 4.0616 R ² : -0.3160 CYP3A4 Accuracy: 0.7518 F1: 0.6967 AUROC: 0.8293 AMES Accuracy: 0.7834 F1: 0.8032 AUROC: 0.8649	Half_Life RMSE: 103.6412 MAE: 27.5290 R ² : 0.2512 Clearance_Hepatocyte RMSE: 47.6597 MAE: 36.7164 R ² : 0.1068 VDss RMSE: 13.3170 MAE: 4.7640 R ² : -2.7858 CYP3A4 Accuracy: 0.7101 F1: 0.5106 AUROC: 0.8297 AMES Accuracy: 0.7442 F1: 0.7500 AUROC: 0.8330	Half_Life RMSE: 115.2806 MAE: 33.9431 R ² : 0.0735 Clearance_Hepatocyte RMSE: 42.8542 MAE: 35.5413 R ² : 0.2608 VDss RMSE: 8.0255 MAE: 6.1927 R ² : -0.3751 CYP3A4 Accuracy: 0.6894 F1: 0.4753 AUROC: 0.7868 AMES Accuracy: 0.7558 F1: 0.7757 AUROC: 0.8320

Raport Ewaluacji Modeli Multi-Task Learning (MTL) dla algorytmu Random Forest

Zestaw Tematyczny: KARDIOTOKSYCZNOSC (hERG, Lipophilicity, Solubility, VDss, AMES)

Zadania MTL	Descriptors	Fingerprints	Embeddings
hERG + Lipophilicity	Lipophilicity RMSE: 0.8587 MAE: 0.6433 R²: 0.5010	Lipophilicity RMSE: 0.8981 MAE: 0.6891 R²: 0.4541	Lipophilicity RMSE: 0.9740 MAE: 0.7729 R²: 0.3579
	hERG Accuracy: 0.8473 F1: 0.9029 AUROC: 0.8217	hERG Accuracy: 0.8397 F1: 0.9014 AUROC: 0.7979	hERG Accuracy: 0.8168 F1: 0.8889 AUROC: 0.8986
hERG + Lipophilicity + Solubility	Lipophilicity RMSE: 0.8683 MAE: 0.6523 R²: 0.4897	Lipophilicity RMSE: 0.9382 MAE: 0.7290 R²: 0.4042	Lipophilicity RMSE: 1.0280 MAE: 0.8236 R²: 0.2847
	Solubility RMSE: 1.1102 MAE: 0.7576 R²: 0.7729	Solubility RMSE: 1.4883 MAE: 1.1405 R²: 0.5918	Solubility RMSE: 1.4295 MAE: 1.1091 R²: 0.6234
	hERG Accuracy: 0.8397 F1: 0.8976 AUROC: 0.8292	hERG Accuracy: 0.8473 F1: 0.9029 AUROC: 0.8112	hERG Accuracy: 0.8321 F1: 0.8972 AUROC: 0.9085
hERG + Lipophilicity + Solubility + VDss	Lipophilicity RMSE: 0.8680 MAE: 0.6576 R²: 0.4901	Lipophilicity RMSE: 1.0126 MAE: 0.7995 R²: 0.3060	Lipophilicity RMSE: 1.1535 MAE: 0.9433 R²: 0.0994
	Solubility RMSE: 1.1216 MAE: 0.7673 R²: 0.7682	Solubility RMSE: 1.6171 MAE: 1.2666 R²: 0.5181	Solubility RMSE: 1.7196 MAE: 1.3794 R²: 0.4551
	VDss RMSE: 5.6664 MAE: 3.3597 R²: 0.3146	VDss RMSE: 9.8217 MAE: 4.2188 R²: -1.0593	VDss RMSE: 7.6017 MAE: 5.3933 R²: -0.2337
	hERG Accuracy: 0.8397 F1: 0.8976 AUROC: 0.8397	hERG Accuracy: 0.8397 F1: 0.8976 AUROC: 0.8319	hERG Accuracy: 0.8168 F1: 0.8899 AUROC: 0.8930

hERG + VDss	VDss RMSE: 5.6447 MAE: 3.2774 R ² : 0.3198	VDss RMSE: 5.8833 MAE: 3.3689 R ² : 0.2611	VDss RMSE: 7.1359 MAE: 4.9395 R ² : -0.0871
	hERG Accuracy: 0.8244 F1: 0.8878 AUROC: 0.8242	hERG Accuracy: 0.8397 F1: 0.9005 AUROC: 0.8211	hERG Accuracy: 0.8244 F1: 0.8930 AUROC: 0.8949
hERG + Solubility	Solubility RMSE: 1.1088 MAE: 0.7540 R ² : 0.7735	Solubility RMSE: 1.4718 MAE: 1.1261 R ² : 0.6008	Solubility RMSE: 1.4024 MAE: 1.0832 R ² : 0.6376
	hERG Accuracy: 0.8321 F1: 0.8922 AUROC: 0.8227	hERG Accuracy: 0.7710 F1: 0.8469 AUROC: 0.7948	hERG Accuracy: 0.8550 F1: 0.9100 AUROC: 0.8933
Solubility + VDss	Solubility RMSE: 1.1132 MAE: 0.7632 R ² : 0.7717	Solubility RMSE: 1.5856 MAE: 1.2337 R ² : 0.5367	Solubility RMSE: 1.6797 MAE: 1.3386 R ² : 0.4801
	VDss RMSE: 5.6158 MAE: 3.3862 R ² : 0.3268	VDss RMSE: 9.6091 MAE: 4.1636 R ² : -0.9711	VDss RMSE: 7.6891 MAE: 5.6559 R ² : -0.2622
hERG + AMES	hERG Accuracy: 0.8473 F1: 0.9029 AUROC: 0.8159	hERG Accuracy: 0.8168 F1: 0.8812 AUROC: 0.8088	hERG Accuracy: 0.8397 F1: 0.9014 AUROC: 0.8825
	AMES Accuracy: 0.7923 F1: 0.8108 AUROC: 0.8740	AMES Accuracy: 0.8281 F1: 0.8399 AUROC: 0.9034	AMES Accuracy: 0.8074 F1: 0.8248 AUROC: 0.8748

**hERG +
Lipophilicity +
Solubility +
VDss +
AMES**

Lipophilicity

RMSE: 0.8707
MAE: 0.6596
R²: 0.4868

Solubility

RMSE: 1.1224
MAE: 0.7650
R²: 0.7678

VDss

RMSE: 5.6500
MAE: 3.3237
R²: 0.3185

hERG

Accuracy: 0.8626
F1: 0.9118
AUROC: 0.8415

AMES

Accuracy: 0.7730
F1: 0.7945
AUROC: 0.8641

Lipophilicity

RMSE: 1.0213
MAE: 0.8092
R²: 0.2940

Solubility

RMSE: 1.6299
MAE: 1.2768
R²: 0.5104

VDss

RMSE: 12.0398
MAE: 4.5763
R²: -2.0944

hERG

Accuracy: 0.8321
F1: 0.8922
AUROC: 0.8323

AMES

Accuracy: 0.7256
F1: 0.7011
AUROC: 0.8452

Lipophilicity

RMSE: 1.1616
MAE: 0.9525
R²: 0.0867

Solubility

RMSE: 1.7304
MAE: 1.3889
R²: 0.4482

VDss

RMSE: 7.7547
MAE: 5.5379
R²: -0.2838

hERG

Accuracy: 0.8168
F1: 0.8889
AUROC: 0.9054

AMES

Accuracy: 0.7249
F1: 0.7072
AUROC: 0.8276